

Create Cross-platform Mobile Apps Using FOSS Tools

The mantra 'Write once, deploy anywhere' is the principle underlying the development of cross-platform mobile applications. Due to the high demand for mobile apps for all sorts of ventures, it makes plain economic sense to write code once for all platforms and all devices, rather than maintain separate development teams for each. The two FOSS mobile application development frameworks – Apache Cordova and Corona – discussed in this article are designed to help developers do that.



Today, most corporate or government services are deployed on the Web or on the cloud. Websites, mobile apps and cloud based services are launched in cyber space for such implementation, so that the anytime-anywhere access to the services can be made available.

India is one of the biggest and most lucrative markets for smartphone manufacturers. Research analyses and reports from *Statista.com* have shown that India is one of the countries with the biggest market for mobile phones. In 2017, around 134 million smartphones were sold in India. Another research report by *Statista.com* predicts that smartphone users in India will increase to around 450 million by 2022.

There are more than 20 mobile operating systems that can be used in smartphones today, each with their own set of libraries and apps. An organisation wanting a competitive edge obviously needs to develop a mobile app for all these operating systems, so that mobile based services are available and can be accessed on all types of devices and platforms.

The major challenge associated with mobile app development is how to cope with the compatibility issues that come up with so many platforms and devices in use today.

Table 1: Thirty platforms and development environments for mobile apps

Android	Amazfit OS	Bada
BlackBerry	ColorOS	Cyanogen OS
Fire OS	Firefox OS	FunTouch
FlymeOS	Fuschia	iOS
KaiOS	LiteOS	Lune
MIUI	Maemo	MeeGo
Nokia Asha platform	Palm	Pixel UI
PureOS	QNX	Sailfish
Symbian Mobile	Tizen	Ubuntu Touch
WebOS	Windows Phone	Yun

Smartphone apps can be classified into three categories:

- Native apps
- Cross-platform apps